



Final assignment
Microsystems Design and Modelling (ET4260)
Part 3/3

Instructions. Document your calculations properly with derivations and screenshots when needed. Latex-generated PDF files are accepted.

Deadline for final report submission (parts 1+2+3): 24th June 2026.
Corrections made after the first submission are marked in [blue](#).

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1 Part a) Optimization of the phase shifter for quantum applications

The technology now allows a minimum feature size of 200 nm (all widths and gaps), a maximum feature size of 1 mm and $2 < N < 200$; the suspended thickness stays $t = 200$ nm in part a. Throughout this part the lumped-element model of Part 1 is used, with two refinements learned from the Part-2 FEM: the figure-faithful mass bookkeeping (two $L \times W_m$ plates, $N/2$ moving fingers) and the side (lateral) pull-in channels that the FEM identified as the real instability of the old design.

a.1 – Low-frequency modulation sensitivity and V_{DC_MAX}

From Part 1 (Q8), the low-frequency limit of the modulation transfer function is

$$\left| \frac{\Delta\phi}{V_{AC}} \right|_{f \rightarrow 0} (V_{DC}) = \frac{S_\phi(g_{op}) \eta(V_{DC})}{k}, \quad \eta = V_{DC} \frac{N\varepsilon_0 t}{h}, \quad S_\phi = \frac{2\pi L}{\lambda} \Delta\eta_{\text{eff}} b e^{-b g_{op}}, \quad (1)$$

with $\Delta\eta_{\text{eff}} = 0.2$. The comb drive pulls the shuttle *into* the comb, i.e. away from the waveguide (sign verified by the Part-2 FEM), so the operating gap grows with bias:

$$g_{op} = g_0 + x(V_{DC}), \quad x(V_{DC}) = \frac{V_{DC}^2}{2k} \frac{N\varepsilon_0 t}{h} \equiv \alpha V_{DC}^2. \quad (2)$$

The sensitivity is therefore a competition between the electromechanical coupling, linear in V_{DC} , and the exponentially decaying optical overlap:

$$\left| \frac{\Delta\phi}{V_{AC}} \right|_{f \rightarrow 0} = \frac{2\pi L}{\lambda} \Delta\eta_{\text{eff}} b e^{-b(g_0 + \alpha V_{DC}^2)} \frac{N\varepsilon_0 t}{k h} V_{DC}. \quad (3)$$

Setting $d|H|/dV_{DC} = 0$ gives $1 - 2b\alpha V_{DC}^2 = 0$, hence

$$V_{DC_MAX} = \frac{1}{\sqrt{2b\alpha}} = \sqrt{\frac{k h}{b N \varepsilon_0 t}} \quad (4)$$

Two corollaries are worth stating explicitly. First, the displacement at the optimum,

$$x(V_{DC_MAX}) = \alpha V_{DC_MAX}^2 = \frac{1}{2b} = 1.67 \mu\text{m}, \quad (5)$$

is a *purely optical* quantity – independent of every geometric parameter. Whatever the design, maximum sensitivity is reached when the bar has moved $1/(2b)$ away from the waveguide. Second, substituting Eq. 4 back into Eq. 3 gives the closed form

$$\left| \frac{\Delta\phi}{V_{AC}} \right|_{\text{max}} = \frac{2\pi L}{\lambda} \Delta\eta_{\text{eff}} \frac{e^{-b g_0 - 1/2}}{V_{DC_MAX}}, \quad (6)$$

so the achievable peak sensitivity is set by the interaction length L , the rest gap g_0 and – inversely – by V_{DC_MAX} itself: a good design makes kh/N *small*. For the Part-1 dimensions, Eq. 4 gives $V_{DC_MAX} = 1377$ V, far above the derived pull-in voltage (347 V analytically, 222 V in the Part-2 FEM): the sensitivity peak of the old design is unreachable (Fig. 1, left), which is precisely the problem this part fixes.

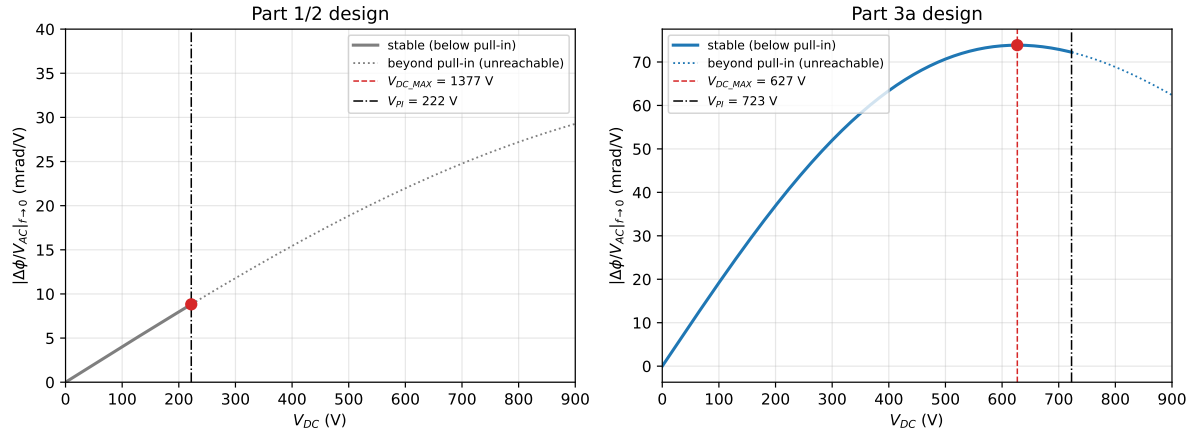


Figure 1: Low-frequency modulation sensitivity vs DC bias, Eq. 3. Left: Part-1/2 design – the peak at 1377 V lies far beyond pull-in (222 V, Part-2 FEM), so only the linear toe of the curve is usable (8.7 mrad/V at best). Right: optimized design – the peak of 73.9 mrad/V at $V_{DC_MAX} = 627$ V lies *below* pull-in (723 V) and is reachable.

a.2 – Conditions for $V_{DC_MAX} < V_{PI}$

With the finger-tip pull-in model of Part 1 (Q7), where u is the tip gap at pull-in and $g_{t0} = L_c - d_0$ the tip clearance at rest,

$$V_{PI} = \sqrt{\frac{k u^3}{N \varepsilon_0 W_c t}}, \quad g_{t0} = \frac{3u}{2} + \frac{u^3}{2hW_c}, \quad (7)$$

the ratio with Eq. 4 is

$$\left(\frac{V_{DC_MAX}}{V_{PI}}\right)^2 = \frac{h W_c}{b u^3} \quad \Rightarrow \quad \boxed{V_{DC_MAX} < V_{PI} \iff b u^3 > h W_c.} \quad (8)$$

The spring constant k , the comb size N , the thickness t and ε_0 all cancel. The condition is purely geometric-optical: stiffening the suspension or adding fingers changes both voltages by the same factor and cannot make the peak reachable. Because $x(V)$ is monotone in V , Eq. 8 has an equivalent and more intuitive displacement form: the *stable comb travel* must exceed the optical optimum of Eq. 5,

$$x_{PI} = g_{t0} - u > \frac{1}{2b} = 1.67 \mu\text{m}. \quad (9)$$

In the deep-clearance limit ($u^3/2hW_c \gg 3u/2$, so $u^3 \rightarrow 2hW_c g_{t0}$) the condition collapses to simply $g_{t0} > 1/(2b)$. The critical terms are therefore:

- **tip clearance** $g_{t0} = L_c - d_0$: must be much larger than $1/(2b)$. Part 1 had $g_{t0} = 1 \mu\text{m}$, giving $x_{PI} = 0.40 \mu\text{m} \ll 1.67 \mu\text{m}$ – the fundamental flaw of the old comb. The redesign needs $g_{t0} \gtrsim 4 \mu\text{m}$;
- **the product hW_c** : it scales the destabilizing tip force; smaller hW_c pushes pull-in deeper into the travel (Eq. 7);
- the optical constant b fixes both sides of the inequality and is not a design variable.

Figure 2 visualizes Eq. 9 for both designs.

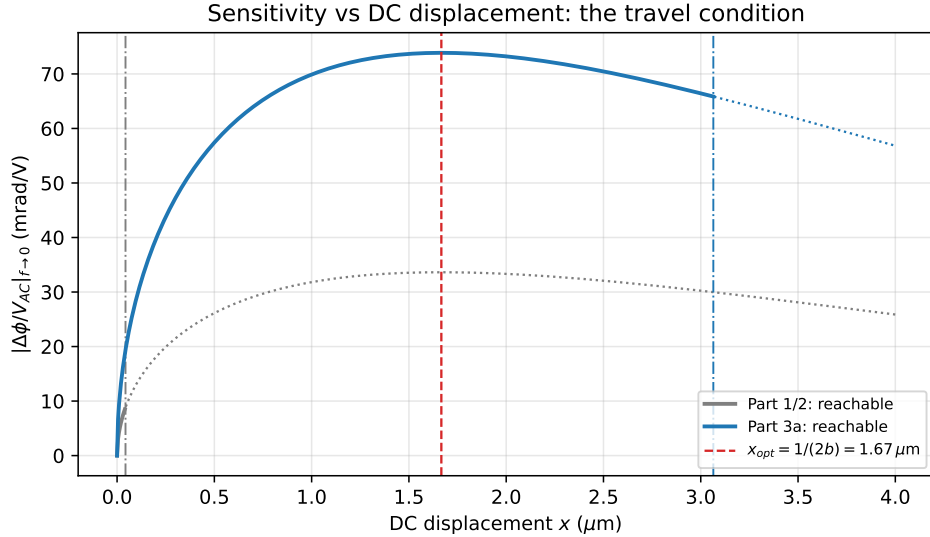


Figure 2: Sensitivity as a function of the DC displacement x . Both curves peak at the universal optimum $x_{opt} = 1/(2b) = 1.67 \mu\text{m}$ (red dashed). The reachable travel of the Part-1/2 design (grey dash-dotted, $\approx 0.04 \mu\text{m}$ at its FEM pull-in) ends long before the optimum; the optimized comb provides $x_{PI} = 3.07 \mu\text{m}$ of stable travel (blue dash-dotted).

Side pull-in: the constraint the 1-DOF model misses. Part 2 showed that the old design actually fails by lateral *rocking* at 222 V, not by the axial tip mechanism at 347 V. Any redesign must therefore also respect the side-instability channels. Four are modelled analytically, all driven by the per-finger lateral negative stiffness $k_{neg} = 2\varepsilon_0 t d_{op} V^2 / h^3$ at the operating engagement $d_{op} = d_0 + 1/(2b)$: (1) individual finger bending (Rayleigh quotient of the engaged cantilever); (2) shuttle rocking against the beam-pair axial stiffness, with column bending in series; (3) *spine bending*: the comb spine is a beam on a negative elastic foundation $\kappa = k_{neg}/p$ (p = comb pitch), whose cantilever half of length ℓ buckles at $\kappa_{crit} = (1.875/\ell)^4 EI_{spine}$ – this is what forbids arbitrarily slim plates; and (4) lateral shuttle translation. Combined à la Dunkerley and calibrated once against the Part-2 FEM (model 247 V vs FEM 222 V $\rightarrow$ factor 0.90), the model reproduces the old design’s side pull-in within 1%, and is used as a constraint set in the optimization below.

a.3 – Optimized dimensions

Formulation. Maximize the achievable peak sensitivity, Eq. 6, $|H|_{max} \propto L e^{-bg_0} / V_{DC_MAX}$, over all eleven geometric parameters, subject to

- the four assignment constraints: $f_r > 500 \text{ kHz}$, $V_{PI} > 200 \text{ V}$, $V_{DC_MAX} < V_{PI}$, $a_{0,max}(\omega_r) > 2 \times 10^4 \text{ m/s}^2$;
- the travel condition Eq. 9 with 20% margin;
- the four side-stability channels (a.2) with 15% voltage margin over V_{DC_MAX} ;
- internal-mode separation: bar/spine tip, finger and beam bending modes $\geq 2.5f_r$ (so that the first eigenmode is cleanly the actuation mode and the rigid-shuttle assumption holds);
- the comb fitting under the spine, $\approx N(W_c + h) + 6W_c \leq L$;
- technology: all features in $[200 \text{ nm}, 1 \text{ mm}]$, $2 < N < 200$, $t = 200 \text{ nm}$ fixed.

The model is the Part-1 lumped element set with figure-faithful masses (validated against the Part-2 FEM in Q1/Q2 there), Q from the Part-1 damping model ($b_{air} = c_{air} A_m$, $Q_{clamp} =$

500), and a full grid search over $\sim 10^6$ candidates followed by rounding to clean values (`analysis/p3a_optimize.py`).

Design logic. The search converges onto a clear logic:

1. g_0 stays at the 200 nm minimum (sensitivity $\propto e^{-bg_0}$);
2. the suspension is softened exactly to the resonance floor – $V_{DC_MAX} \propto \sqrt{kh/N}$, and k may drop as far as $f_r > 500$ kHz allows. Conveniently, at $g_0 = 200$ nm the acceleration constraint $a_{0,\max}(\omega_r) = g_0\omega_r^2/Q$ with $Q \approx \omega_r\rho t/c_{\text{air}} \approx 80$ is satisfied at almost exactly the same frequency, so both floors coincide;
3. the comb is completely re-proportioned: short, stubby fingers ($L_c \times W_c = 5.5 \times 0.7$ μm , internal mode 32 MHz) at a small rest overlap $d_0 = 1$ μm leave a tip clearance $g_{t0} = 4.5$ μm and a stable travel $x_{\text{PI}} = 3.07$ μm , comfortably beyond the 1.67 μm optimum; $N = 58$ gaps at $h = 0.9$ μm fill the spine length;
4. the plates slim down to $W_m = 10$ μm – the spine-bending (negative elastic foundation) limit; thinner plates would buckle sideways electrostatically far below the operating voltage;
5. L stays at 100 μm : the plate mass required for spine stability grows $\propto L^{7/3}$, so the extra optical length of a longer bar is eaten by the higher V_{DC_MAX} – the objective is nearly flat in L , and the shortest stiff device wins on robustness.

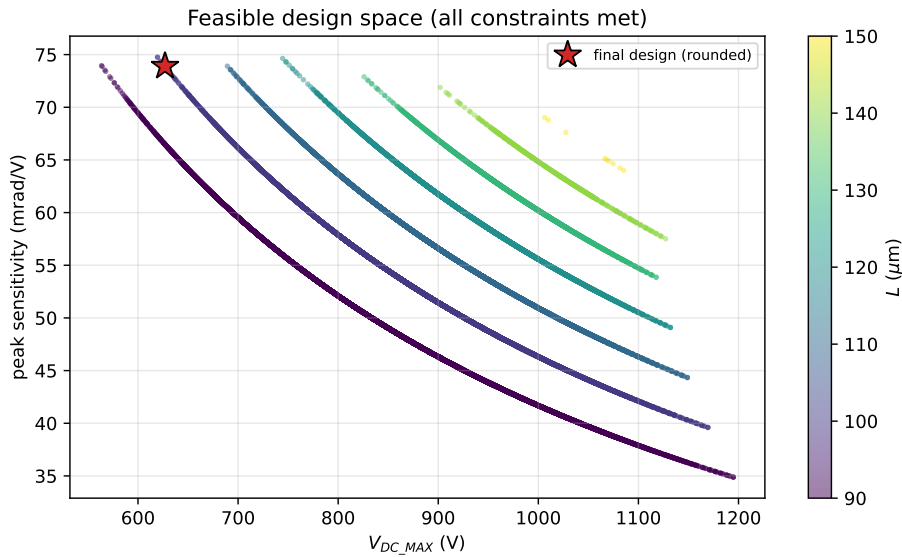


Figure 3: The $\sim 47\,000$ feasible designs of the refined search in the sensitivity– V_{DC_MAX} plane (color: modulation length L). Each arc is one L -family; the objective $|H|_{\max} \propto L/V_{DC_MAX}$ improves toward the top left. The chosen design (star) sits on the frontier; the $\approx 1\%$ gap to the best raw grid point is the cost of rounding to clean dimensions.

Table 1: Final dimensions (the “Final Value” column of the assignment table) next to the Part-1/2 values.

Feature	Symbol	Part 1/2	Part 3a final
Length of modulation section	L	100 μm	100 μm
Modulation gap at rest	g_0	200 nm	200 nm
Length of beams	L_b	75 μm	40 μm
Width of beams	W_b	5 μm	1.85 μm
Length of mass central section	L_m	40 μm	30 μm
Width of mass sections	W_m	30 μm	10 μm
Comb-drive gap	h	1 μm	0.9 μm
Comb-drive overlap at rest	d_0	19 μm	1.0 μm
Comb-drive finger length	L_c	20 μm	5.5 μm
Comb-drive finger width	W_c	1 μm	0.7 μm
Bottom end fingers	$3W_c$	3 μm	2.1 μm
Thickness of suspended parts	t	200 nm	200 nm (fixed)
Number of comb gaps	N	40	58

Table 2: Predicted performance (analytical model) and constraint check.

quantity	Part 1/2	Part 3a	constraint
k	40.3 N/m	13.5 N/m	–
m_{eff}	3.80 pg	1.18 pg	–
f_r	518 kHz (FEM 496)	539 kHz (FEM 522)	$> 500 \text{ kHz } \checkmark$
V_{DC_MAX}	1377 V	627 V	$< V_{PI} \checkmark$ (margin 1.15)
V_{PI} axial / side	347 / 222 V	744 / 723 V	$> 200 \text{ V } \checkmark$
x_{PI}	0.40 μm	3.07 μm	$> 1.67 \mu\text{m } \checkmark$
$a_{0,\text{max}}(\omega_r), Q$	$2.7 \times 10^4 \text{ m/s}^2, 77$	$2.8 \times 10^4 \text{ m/s}^2, 81$	$> 2 \times 10^4 \text{ m/s}^2 \checkmark$
peak $ \Delta\phi/V_{AC} _{f \rightarrow 0}$ (achievable)	8.7 mrad/V (222 V)	73.9 mrad/V (627 V)	maximized

The achievable sensitivity improves by $8.5\times$ over the old design’s best stable operating point, and by $183\times$ over the 10 V operating point quoted in Part 1. One practical remark: 627 V across a 0.9 μm gap is a $7 \times 10^8 \text{ V/m}$ field – acceptable for the implied vacuum/cryogenic quantum application, but it requires vacuum packaging (in air, breakdown would intervene first). The voltage is high because the fixed $t = 200 \text{ nm}$ keeps the actuator capacitance small; the thicker device layer of part b would also lower $V_{DC_MAX} \propto 1/\sqrt{t}$.

a.4 – COMSOL validation of the new design

The new geometry was rebuilt in COMSOL 6.4 with the same modelling recipe validated in Part 2: 2D plane stress with $t = 200 \text{ nm}$, isotropic Si ($E = 170 \text{ GPa}$, $\nu = 0.28$, $\rho = 2330 \text{ kg/m}^3$), fixed beam anchors; for the electromechanical studies the shuttle is a Domain Terminal at 0 V, the stator floor and fixed-finger walls carry V_d , an Electromechanical Forces coupling acts on the solids and the air cavity deforms with a hyperelastic moving mesh. Figure 4 shows the mesh.

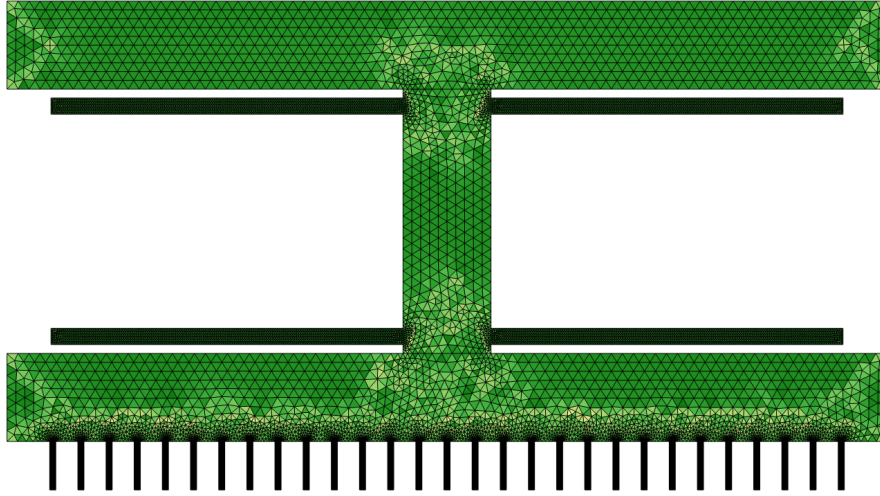


Figure 4: Mesh of the mechanical model (normal refinement, 12 731 elements). The element size is refined in the comb region where features are 0.7–0.9 μm .

(i) Equivalent spring constant. A 1 μN test force on the spine along $-y$ gives $k = F/|u_y|$ (Table 3). The FEM value, $k = 12.50 \text{ N/m}$, is 7.1% below the analytical $4EtW_b^3/L_b^3 = 13.45 \text{ N/m}$ – the same plate-compliance and shear effects quantified in Part 2 (there –8.7%). The deviation is smaller here because the slimmer plates contribute less of the total compliance budget relative to the much softer beams. The design anticipated this with the Part-2-calibrated factor 0.92 ($\rightarrow 12.38 \text{ N/m}$, within 1% of the FEM).

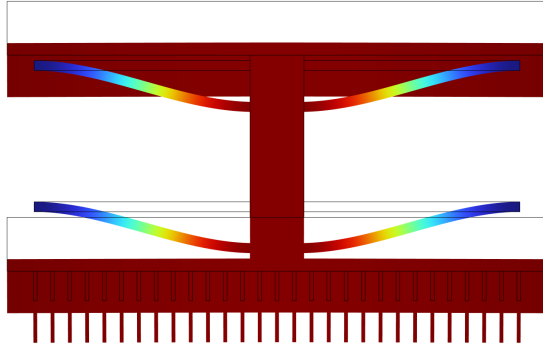
Table 3: Mesh convergence of the spring constant.

mesh	elements	k (N/m)
coarse	4 429	12.578
normal	12 731	12.525
fine	40 916	12.501

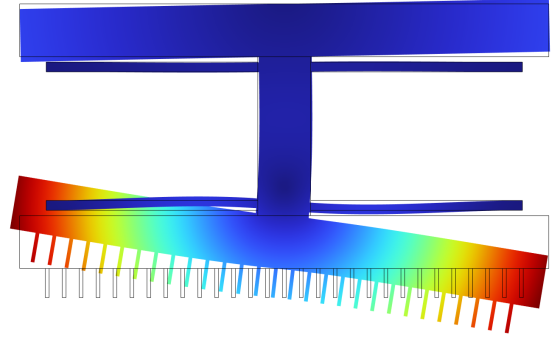
(ii) First five eigenmodes. Table 4 and Fig. 5 summarize the eigenfrequency analysis (frequencies converged to 0.1% between two mesh refinements). The fundamental mode at 521.7 kHz satisfies $f_r > 500 \text{ kHz}$ $\checkmark$. The analytical prediction (539 kHz) is 3.2% high – exactly the softer FEM spring, $\sqrt{12.50/13.45} = 0.964$; the FEM modal mass $k/\omega_1^2 = 1.163 \text{ pg}$ matches the lumped estimate (1.175 pg) to 1%. Unlike Part 2, where the comb fingers formed a band at 3.3 MHz, the redesigned stubby fingers have internal modes near 32 MHz and disappear from the low spectrum entirely; modes 2–5 are plate rotation/bending modes, the family the side-stability constraints deliberately keep stiff. The separation $f_2/f_1 = 5.1$ means the device behaves as a clean single-mode actuator over a wide band.

Table 4: First five eigenfrequencies (fine mesh).

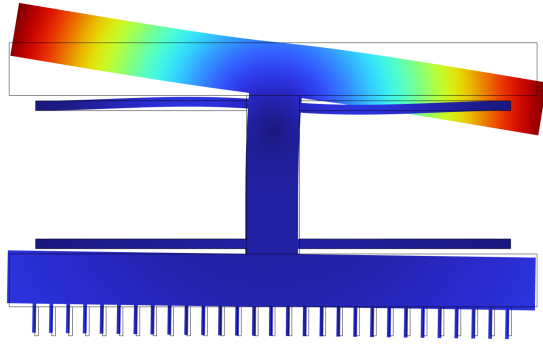
mode	character	f (kHz)
1	shuttle translation along y (actuation mode)	521.7
2	spine in-plane rotation, fingers swinging laterally	2634.8
3	bar in-plane rotation (symmetric partner of mode 2)	2786.1
4	symmetric in-plane bending of the plate tips	5519.3
5	higher-order plate bending with beam second modes	7616.5



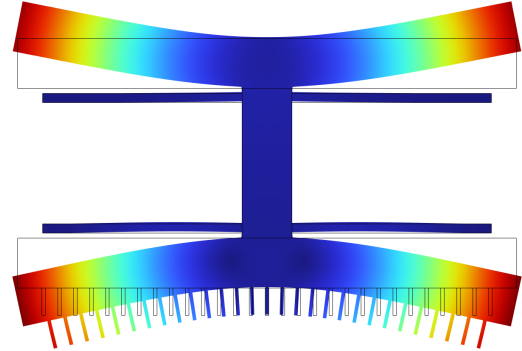
(a) mode 1: 521.7 kHz



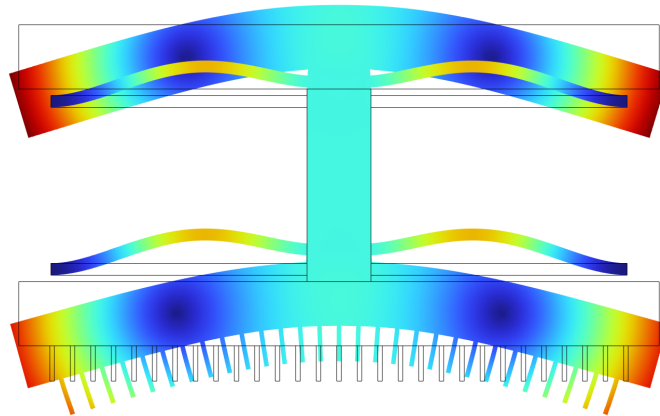
(b) mode 2: 2634.8 kHz



(c) mode 3: 2786.1 kHz



(d) mode 4: 5519.3 kHz



(e) mode 5: 7616.5 kHz

Figure 5: 2D modal displacement maps of the first five modes of the optimized design.

(iii) **Maximum DC displacement at the intended operating voltage.** A stationary continuation sweep $V_d = 0 \rightarrow 760$ V (steps of 10 V, 627 V included) tracks the stable branch

(Fig. 6, left). At the design operating voltage,

$$x(627 \text{ V})_{\text{FEM}} = 0.919 \mu\text{m} \quad \text{toward the comb (away from the waveguide),}$$

vs the analytical optimum $1/(2b) = 1.667 \mu\text{m}$ – 45% below the lumped prediction. The sweep data identify two physical causes that the ideal-comb model misses:

1. *Shallow-engagement (fringe-regime) comb constant.* The force-implied $(dC/dx)_{\text{eff}} = 2k_{\text{FEM}}|u_y|/V^2$ is only $\approx 71 \text{ pF/m}$ at low bias, vs the ideal $N\varepsilon_0 t/h = 114 \text{ pF/m}$ (Fig. 6, right). With $d_0 = 1 \mu\text{m}$ the rest overlap is only $1.1h$: the parallel-plate flank field is not yet developed, and the constant- dC/dx assumption only becomes valid at engagements of a few h . (Part 2 never exposed this – its $d_0 = 19 \mu\text{m}$ comb was deeply engaged.)
2. *Beam stress stiffening.* At micron stroke the fixed-guided beams stretch ($x/W_b \approx 0.5$ – 0.9); the prestressed eigenfrequencies (Fig. 9) show f_1 rising from 522 kHz at 0 V to 586 kHz at 450 V – a +26% tangent-stiffness increase that further suppresses dx/dV .

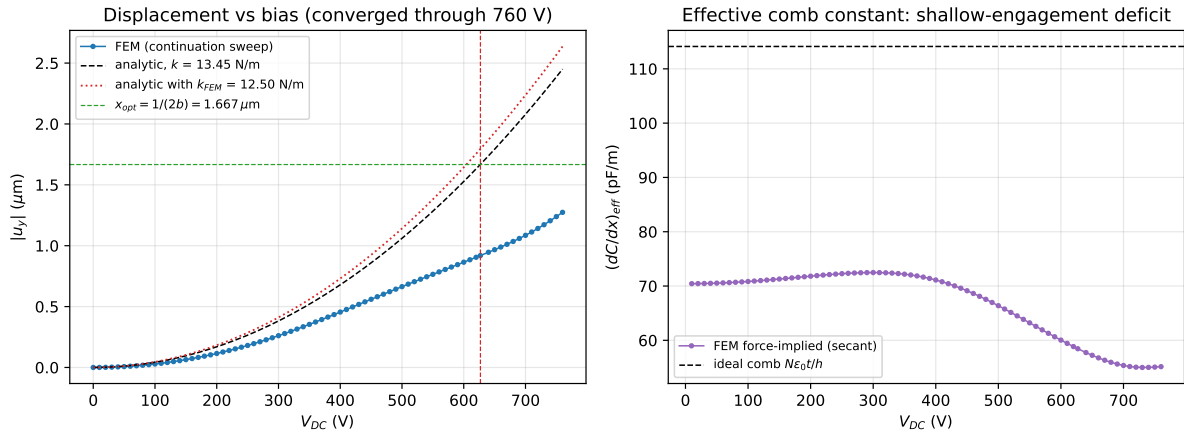


Figure 6: Left: displacement vs bias – FEM continuation sweep against the analytical curve (with both k_{ana} and k_{FEM}). The sweep converges through 760 V with no fold: no pull-in in the operating range. Right: the force-implied effective comb constant, showing the shallow-engagement deficit relative to the ideal $N\varepsilon_0 t/h$.

The consequence is a shift, not a loss: the FEM-derived low-frequency sensitivity $|H|(V) = S_\phi(g_0 + x) dx/dV$ (Fig. 7) reaches 35.7 mrad/V at the design’s 627 V and 54.9 mrad/V at 760 V, still rising – the optimum displacement $1/(2b)$ is reached near 870 V by extrapolation. Because $S(x) \propto \sqrt{x} e^{-bx}$ is very flat around its maximum, operating anywhere on the 750–900 V plateau recovers the designed sensitivity to within a few percent, and the redesign goal – a reachable sensitivity maximum – is preserved, since pull-in moves up even faster (see the pull-in verification below).

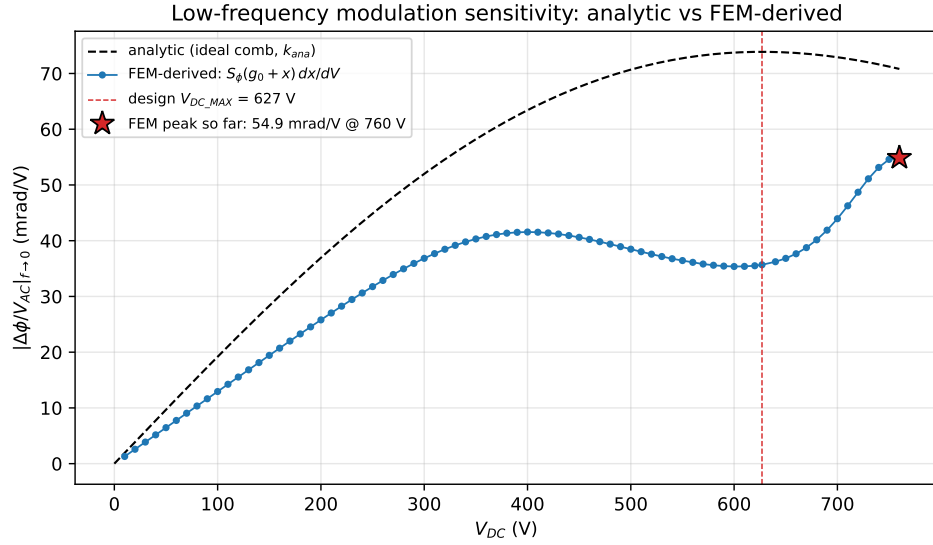


Figure 7: Low-frequency modulation sensitivity vs bias: analytical (ideal comb) and FEM-derived from the static sweep, $S_\phi(g_0 + x) dx/dV$. The three FEM regimes are visible: the fringe-deficit linear rise, the stress-stiffening plateau around 400–600 V, and the recovery at deep engagement. The FEM optimum lies near 870 V, beyond the design value but still far below pull-in.

(iv) Maximum DC acceleration without hitting the waveguide. A body load ρa on all suspended silicon (+y, toward the waveguide) at $V_d = 0$ (worst case – bias only adds clearance) gives contact, $u_{y,\text{bar}} = +g_0$, at

$$a_{\text{max,DC}} = 2.12 \times 10^6 \text{ m/s}^2 = 2.16 \times 10^5 g$$

(verification solve at a_{max} : residual -0.6% , the small deficit being the onset of geometric stiffening). The lumped check $k_{\text{FEM}}g_0/m_{\text{static}} = 2.10 \times 10^6 \text{ m/s}^2$ agrees to 1.3% ; the fully analytical value (k_{ana}) is 6% high, again the spring softening. Note that this quasi-static number is not the binding requirement: the design constraint was the worst-case-frequency AC value $a_{0,\text{max}}(\omega_r) = g_0\omega_r^2/Q = 2.8 \times 10^4 \text{ m/s}^2 > 2 \times 10^4 \text{ m/s}^2 \checkmark$, which is $Q \approx 81$ times smaller than the DC limit because resonance amplifies the base excitation.

(v) Step response and rise time. Rayleigh damping is set to stiffness-proportional, $\beta_{dK} = 1/(2\pi f_1 Q)$ with the analytical $Q = 80.7$ at the FEM $f_1 = 521.7 \text{ kHz}$, and a $1 \mu\text{N}$ force step is applied to the spine from rest (Fig. 8). The final displacement, 79.7 nm , matches the static F/k_{FEM} to 0.4% . The 10–90% **rise time** is $0.30 \mu\text{s}$ – the force step accelerates the 1.18 pg shuttle within a fraction of the $1.92 \mu\text{s}$ oscillation period, in agreement with the undamped $1 - \cos \omega_1 t$ first-rise estimate $1.02/\omega_1 = 0.31 \mu\text{s}$. Because the device is strongly underdamped ($\zeta = 1/2Q = 6.2 \times 10^{-3}$), the response then rings at f_1 and **settles to 5% in** $148.6 \mu\text{s}$, in agreement with the analytical envelope estimate $t_s = \ln(20)/(\zeta\omega_1) = 147.5 \mu\text{s}$ (0.7%). Compared with Part 2 ($158 \mu\text{s}$ at $f_1 = 496 \text{ kHz}$, $Q = 83$): $t_s \propto Q/f_1$, so the slightly faster, slightly less damped new device settles marginally sooner. For the modulator application the rise time is the relevant number; the long ring-down only matters as a recovery time after shocks or bias steps.

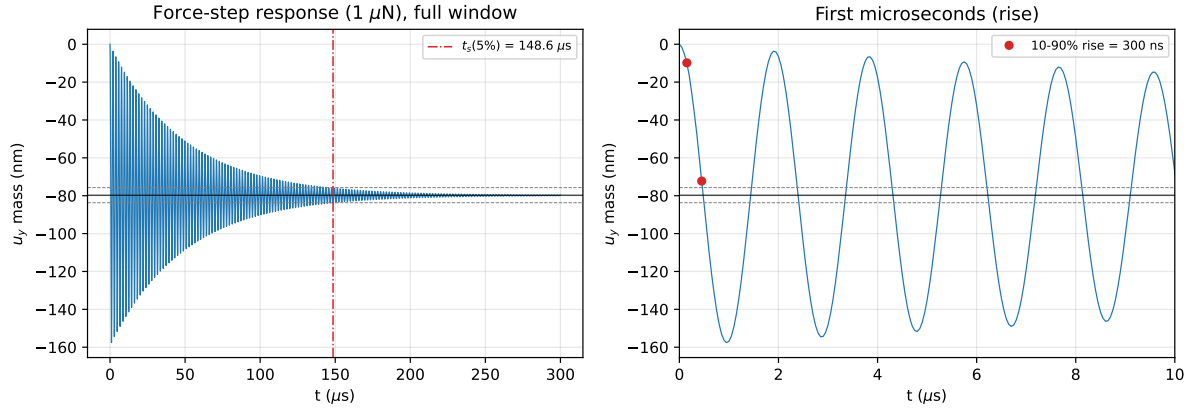


Figure 8: Force-step response of the optimized device (1 μN on the spine, $Q = 80.7$). Left: full window with the $\pm 5\%$ band and settling time. Right: the first ten microseconds with the 10–90% rise markers.

Pull-in verification. Because the whole design hinges on $V_{DC_MAX} < V_{PI}$, two checks beyond the assignment list were run. First, the static continuation sweep converges through 760 V with no fold – no pull-in anywhere in or near the operating range. Second, prestressed eigenfrequencies vs bias (a continuation-ramped stationary solve followed by an eigenfrequency study at each bias) expose which mode loses stiffness (Fig. 9):

Table 5: Prestressed eigenfrequencies vs DC bias.

V_d	f_1 (actuation)	f_2 (spine rotation = side mode)
0 V	522.2 kHz	2636.7 kHz
400 V	557.4 (+6.7%)	2512.7 (−4.7%)
627 V	675.2 (+29%)	2170.5 (−18%)
760 V	584.5	1519.5 (−42%)

f_1 is non-monotonic: stress stiffening dominates up to ≈ 650 V (+29%), then electrostatic softening takes over; $f_1^2 \rightarrow 0$ extrapolates to ≈ 1100 V. The critical channel is mode 2 – the spine-rotation (side) mode – whose $f_2^2 \rightarrow 0$ extrapolation gives $V_{PI}^{FEM} \approx 876$ V. Three conclusions: (i) the analytically designed side pull-in (723 V) is 17% conservative, as expected from the Dunkerley combination; (ii) the *character* of the instability – lateral, through the comb’s negative lateral stiffness, not the axial tip fold – confirms the Part-2 lesson and the constraint set used in the optimization; and (iii) the ordering is as designed, V_{op} (627) $< V_{PI}^{side}$ (≈ 876) $< V_{PI}^{axial}$ (≈ 1100 V), so **all four assignment constraints hold in the FEM**: $f_r = 521.7$ kHz, $V_{PI} \approx 876$ V > 200 V, $V_{DC_MAX} < V_{PI}$ (28% margin), and $a_{0,max}(\omega_r) = 2.8 \times 10^4$ m/s².

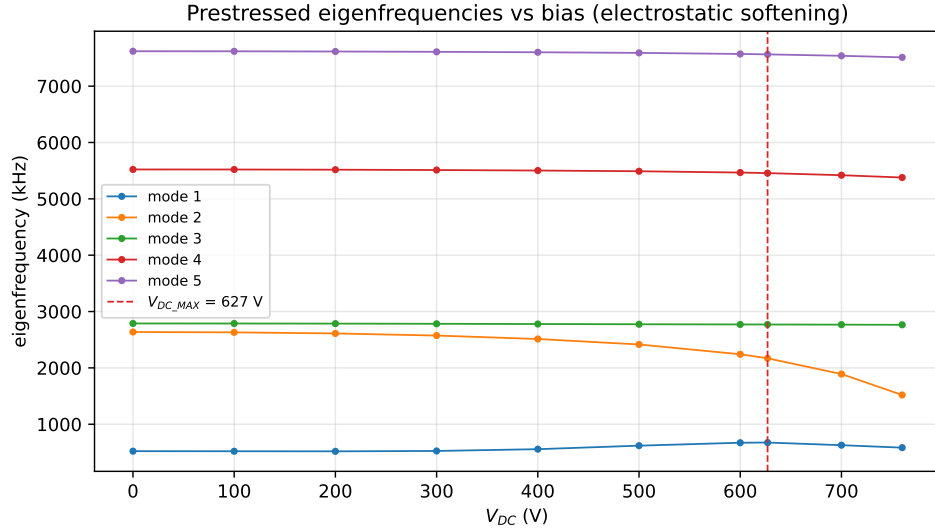


Figure 9: Prestressed eigenfrequencies vs bias. Mode 2 (spine rotation, the side channel guarded by the design constraints) softens toward pull-in at ≈ 876 V; mode 1 first *stiffens* (+29% at the operating point, beam stress stiffening) before electrostatic softening takes over.

Summary of deviations and possible second iteration. Table 6 condenses the analytic $\leftrightarrow$ FEM comparison. The two systematic effects – shallow-engagement comb force and beam stress stiffening – *shift* the sensitivity optimum from 627 V into the 750–900 V range without destabilizing it, because the same stiffening physics raises pull-in even faster. A second design iteration could recover the lower-voltage optimum by deepening the rest overlap ($d_0 \rightarrow 2.5\text{--}3\text{ }\mu\text{m}$, restoring the ideal dC/dx) and lengthening the beams at constant k (reducing x/W_b and the hardening), at a modest cost in side-stability margin; since all assignment constraints are already met, this is left as a documented refinement.

Table 6: Analytical model vs COMSOL: the a.4 comparison at a glance.

quantity	analytic	FEM	cause of deviation
k	13.45 N/m	12.50 N/m (−7%)	plate compliance, shear
f_r	538.6 kHz	521.7 kHz (−3%)	softer k
$x(627\text{ V})$	1.667 μm	0.919 μm (−45%)	fringe-regime dC/dx ($\times 0.62$) + stress stiffening (+26% k_{tan})
peak $ H $	73.9 mrad/V @ 627 V	≈ 55 mrad/V @ 760 V, rising	same two causes; flat $S(x)$ limits the loss
V_{PI} (side)	723 V	≈ 876 V (+17%)	Dunkerley conservatism; stiffening aids stability
$a_{\text{max,DC}}$	2.26×10^6 m/s ²	2.12×10^6 m/s ² (−6%)	softer k
$t_{s, \text{rise}}$	147.5 μs , 0.31 μs	148.6 μs , 0.30 μs	agreement $\leq 1\%$

2 Part b) 3D model to mitigate out-of-plane behavior

This part returns to the **Part-2 dimensions** and removes the comb fingers from the geometry, as instructed.

b.1 – Why $t = 200\text{ nm}$ is problematic

A fixed-guided beam bending in plane has the second moment $I_{\text{ip}} = tW_b^3/12$; bending out of plane simply swaps the section dimensions, $I_{\text{oop}} = W_bt^3/12$. With four beams in parallel,

$$k_{\text{ip}} = \frac{4EtW_b^3}{L_b^3} = 40.3\text{ N/m}, \quad k_{\text{oop}} = \frac{4Et^3W_b}{L_b^3} = k_{\text{ip}} \left(\frac{t}{W_b} \right)^2, \quad (10)$$

$$\frac{k_{\text{oop}}}{k_{\text{ip}}} = \left(\frac{0.2}{5} \right)^2 = \frac{1}{625} \Rightarrow k_{\text{oop}} = 0.064\text{ N/m}. \quad (11)$$

The modal mass is the same in both directions, so $f_{\text{oop}} = f_{\text{ip}} (t/W_b) \approx 496/25 \approx 20\text{ kHz}$. The consequences:

- the *true* fundamental mode of the physical device is out-of-plane flapping at $\sim 20\text{ kHz}$ – not the 500 kHz actuation mode the 2D analyses assumed. Ambient acoustic and structural vibration sits exactly in that band and is amplified by Q ;
- out-of-plane shock tolerance drops by the same factor 625: modest accelerations deflect the bar vertically by hundreds of nanometers, modulating the evanescent coupling to the waveguide (parasitic amplitude/phase noise) and risking stiction to the substrate;
- comb levitation – the vertically asymmetric fringe field of an open comb – directly forces this soft degree of freedom at the drive frequency;
- a 200 nm film spanning $100\text{ }\mu\text{m}$ is also highly susceptible to residual-stress warping.

b.2 – Thickness for an in-plane fundamental mode; 3D eigenfrequency analysis

Both k_{ip} and the mass scale linearly with t , so f_{ip} is *independent of thickness*, while $f_{\text{oop}} \propto t$. The naive (beam-only) criterion for the fundamental mode to become the intended in-plane actuation mode is therefore

$$k_{\text{oop}} \geq k_{\text{ip}} \iff t \geq W_b = 5\text{ }\mu\text{m}, \quad (12)$$

with exact degeneracy at $t = W_b$.

3D eigenfrequency analysis. A 3D model (extruded Part-2 layout without fingers, swept mesh, thickness as a parameter) was solved for $t = 0.2 \dots 12\text{ }\mu\text{m}$ (Fig. 10). Three results stand out.

1. The in-plane actuation mode stays at $512\text{--}515\text{ kHz}$ for *all* thicknesses, confirming its t -independence (FEM 514.8 vs analytical 531 kHz : -3% , the familiar softer FEM spring). At $t = 200\text{ nm}$ the lowest out-of-plane mode is 20.0 kHz – matching the hand estimate $f_{\text{ip}} t/W_b \approx 20\text{ kHz}$ exactly and quantifying the b.1 hazard.
2. **The naive crossover criterion Eq. 12 fails in 3D.** At $t = 5\text{ }\mu\text{m}$ the lowest out-of-plane mode sits at 379.5 kHz , still far below the actuation mode, and even at $t = 6\text{ }\mu\text{m}$ it is lower (430.6 kHz). The reason: the lowest out-of-plane mode is *not* the pure flap described by k_{oop} , but a **see-saw pitch mode** – bar and spine moving antiphase in z , rotating about the beam axis (Fig. 11b). A lumped pitch estimate (four beams' out-of-plane bending stiffness $12EI_{\text{oop}}/L_b^3$ on the beam-level lever arms, against the plate inertia at $\approx 35\text{ }\mu\text{m}$ radius) gives $f_{\text{pitch}} \approx 0.55\text{ MHz}$ at $t = 10\text{ }\mu\text{m}$, vs FEM 608 kHz – beam torsion, neglected in the estimate, accounts for the rest. Because the pitch mode rises more slowly with t than the flap, the true crossover sits at $t \approx 7.5\text{ }\mu\text{m}$.

3. **Final thickness:** $t = 10\ \mu\text{m}$, giving an in-plane fundamental at 514.8 kHz ($> 500\ \text{kHz}$ ✓) with an 18% margin to the first out-of-plane mode – a standard SOI device-layer thickness. All other dimensions keep their Part-2 values.

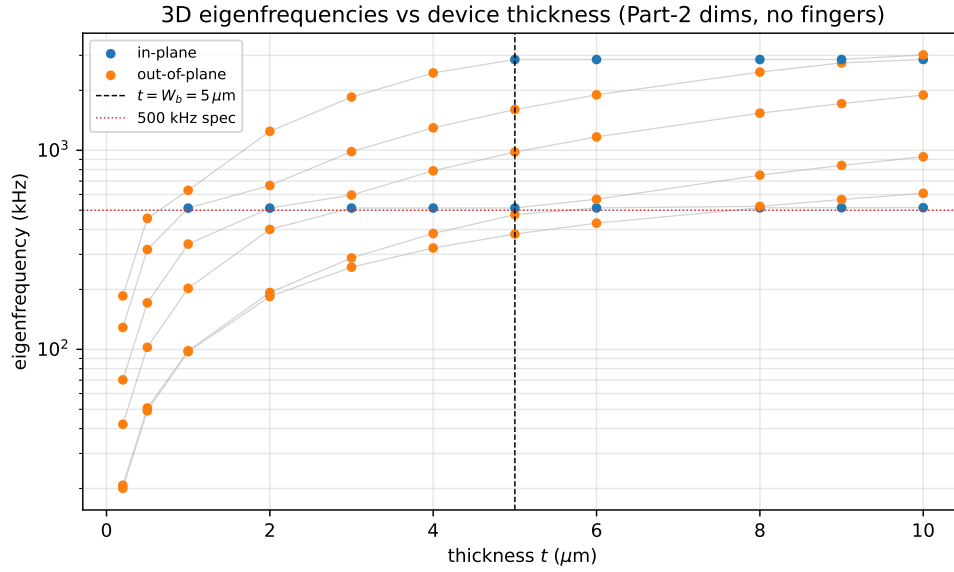


Figure 10: 3D eigenfrequencies vs device thickness (Part-2 dimensions, comb fingers removed). Blue: in-plane modes; orange: out-of-plane modes. The in-plane actuation mode is thickness-independent at $\approx 513\ \text{kHz}$, while the out-of-plane family rises with t ; the fundamental becomes in-plane only at $t \approx 7.5\ \mu\text{m}$ – beyond the naive $t = W_b = 5\ \mu\text{m}$ (dashed) – because the lowest out-of-plane mode is the see-saw pitch, not the flap.

First five 3D eigenmodes and spring constants at $t = 10\ \mu\text{m}$. Table 7 and Fig. 11 report the final modal analysis; Table 8 the equivalent spring constants.

Table 7: First five 3D eigenmodes at $t = 10\ \mu\text{m}$.

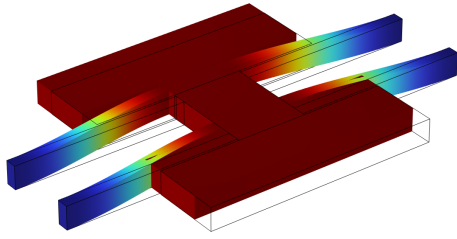
mode	character	f (kHz)
1	in-plane y-translation (actuation) – now fundamental	514.8
2	out-of-plane see-saw pitch (bar/spine antiphase)	607.9
3	out-of-plane flap (shuttle translates in z)	927.9
4	higher out-of-plane (antisymmetric beam-pair/plate bending)	1 894.1
5	in-plane shuttle rotation (t -independent family)	2 858.9

Table 8: Equivalent spring constants at $t = 10\ \mu\text{m}$ (1 μN on the spine plate).

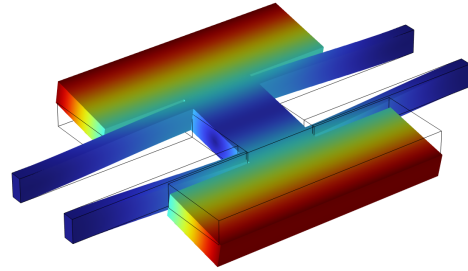
direction	FEM	analytic	deviation
in-plane y	1883 N/m	2015 N/m	−6.5%
out-of-plane z (static)	1641 N/m	8059 N/m	−80%

The in-plane value shows the usual -6.5% plate-compliance/shear softening. The large out-of-plane discrepancy is the same physics as the mode ordering: a static z -force applied at the spine is offset from the pitch axis, so the measured compliance is dominated by the *pitch* degree of freedom, not the flap that $4Et^3W_b/L_b^3$ describes. A modal cross-check on the flap mode itself behaves: $f_{\text{flap,FEM}} = 927.9\ \text{kHz}$ vs $f_{\text{ip}}(t/W_b) = 1030\ \text{kHz}$ (-10% , beam-end rotation and shear at the now-stubby $t/L_b = 0.13$ beam aspect). The general lesson: in 3D the “out-of-plane spring

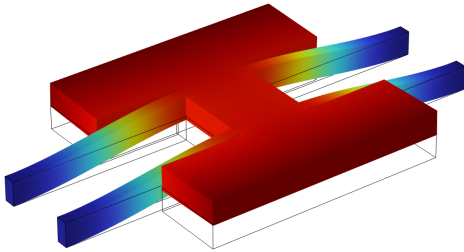
constant” is mode-specific, and for a two-plate shuttle the lowest out-of-plane mode is rotational, which only a 3D model (or a multi-DOF lumped model) can expose.



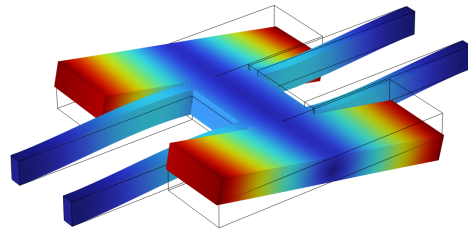
(a) mode 1: 514.8 kHz, in-plane actuation



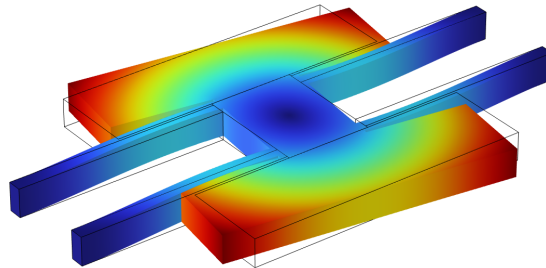
(b) mode 2: 607.9 kHz, see-saw pitch



(c) mode 3: 927.9 kHz, flap



(d) mode 4: 1894.1 kHz



(e) mode 5: 2858.9 kHz, in-plane rotation

Figure 11: First five 3D eigenmodes at the final thickness $t = 10\ \mu\text{m}$ (Part-2 dimensions, comb fingers removed).